

Volatility Squeeze Indicator

thinkScript Indicator Report

Prepared by: Bushra

What This Is

This report documents a custom **Volatility Squeeze** indicator built using thinkScript, the scripting language used by the thinkorswim trading platform. The indicator detects periods when market volatility contracts sharply — price consolidating into a tight range — by comparing Bollinger Bands against Keltner Channels. When the Bollinger Bands move inside the Keltner Channels, it signals that volatility has compressed and a breakout move is likely coming soon.

Tools Used

Tool / Platform	Purpose
thinkorswim (desktop)	Charting platform the study runs on
thinkScript Editor	Where the indicator code was written
thinkScript language	Language used to build the custom study
PaperMoney (simulated trading)	Used to test the indicator on a live TSLA chart

Indicator Logic

The script is added as a lower study (declare lower). It computes a moving-average basis line, then builds Bollinger Bands using a standard deviation multiplier (bbMult) and Keltner Channels using an Average True Range multiplier (kcMult). A squeeze is detected when the Bollinger Band width narrows inside the Keltner Channel width, plotted as yellow dots beneath price/volume, with an alert firing the moment a squeeze begins.

Parameters

A table is used here (instead of a paragraph) because the parameters are structured, repeating data — each input has a name, a default value, and a purpose. A table lets these be scanned and compared at a glance, which is the standard way to document indicator settings.

Input	Default Value	Purpose
length	20	Lookback period for moving average / StDev / ATR
bbMult	2.0	Standard deviation multiplier for Bollinger Bands
kcMult	1.5	ATR multiplier for Keltner Channels

thinkScript Code (Full)

```
declare lower;

input length = 20;
input bbMult = 2.0;
input kcMult = 1.5;

def basis = Average(close, length);
def dev = bbMult * StDev(close, length);
def upperBB = basis + dev;
def lowerBB = basis - dev;

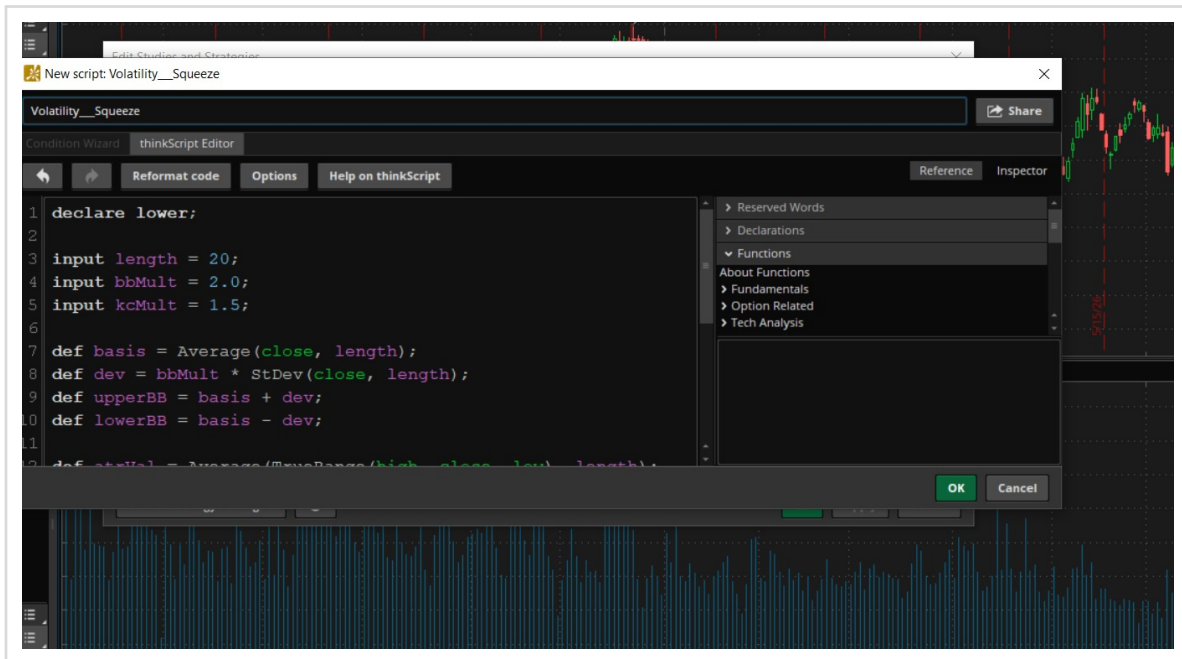
def atrVal = Average(TrueRange(high, close, low), length);
def upperKC = basis + kcMult * atrVal;
def lowerKC = basis - kcMult * atrVal;

def squeeze = (lowerBB > lowerKC) and (upperBB < upperKC);

plot SqueezeOn = if squeeze then 1 else 0;
SqueezeOn.SetPaintingStrategy(PaintingStrategy.BOOLEAN_POINTS);
SqueezeOn.SetDefaultColor(Color.YELLOW);

Alert(squeeze, "Squeeze Detected", Alert.BAR, Sound.Bell);
```

Screenshot — thinkScript Editor



New script "Volatility_Squeeze" open in the thinkScript Editor.

Screenshot — Applied to TSLA Daily Chart



TSLA (1Y, Daily) with the Volatility Squeeze study plotted in the lower pane beneath price and volume. Yellow dots mark squeeze conditions.

Screenshot — Indicator Panel Detail



Full thinkorswim workspace showing the Volatility Squeeze study active on the TSLA chart, with account/watchlist panels for context.